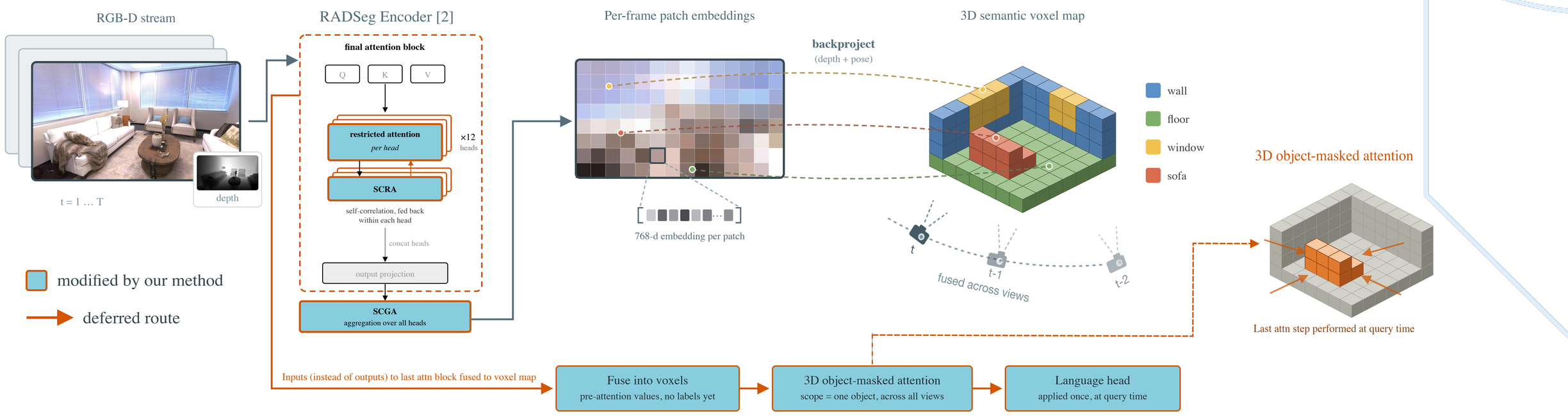
Region-Restricted Attention in 3D Semantic Mapping

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Setting. Semantic voxel mapping labels a scene by encoding each posed RGB-D frame in 2D and fusing the per-patch features into a 3D voxel grid.

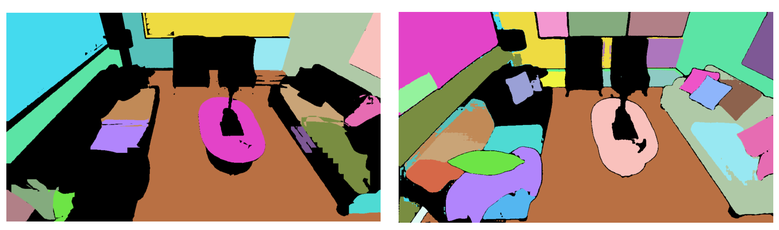
Problem. 2D Segmentation quality declines when the final attention block mixes patches across class boundaries; restricting the scope to same-class patches recovers it [1]. That result is per-frame and purely 2D. A mapping system already has depth, pose, and many views of the same object, and a per-frame scope uses none of them.

Result. We make two changes. *Where the regions come from*: SAM [3] masks, or planar surfaces fitted directly from the depth map, with PDA [4] supporting depth where the sensor is noisy. *Where the attention runs*: per frame (2D), or deferred into the map (3D). Both region sources beat the baseline. Deferring to 3D and scoping the attention results in +5.2 f-mIoU on Replica and +4.9 on ScanNet over baseline.

Methods

Step 1: define regions. We obtain class-agnostic regions from either SAM [3] or depth. For depth, we backproject the depth map and fit planes with CAPE [5]. A patch i is assigned to plane S_k when enough of its support lies within τ of the plane; otherwise it is unassigned $\varnothing$:

$$s(i) \in \{1, \ldots, K\} \cup \{\varnothing\}.$$



The left image masks from depth-guided surfaces; the right image's are from SAM

Step 2: restrict final block attention. We prevent cross-region feature mixing by adding a bias to the final attention logits:

$$\text{Attention} = \text{softmax}\left(\frac{QK^\top}{\sqrt{d}} + B\right) V,$$

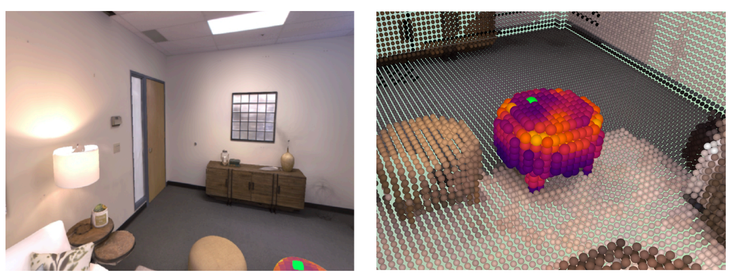
$$B_{ij} = \begin{cases} -\infty, & s(i) \neq \varnothing,\ s(j) \neq s(i), \\ -\dfrac{\|p_i - p_j\|^2}{2\sigma^2}, & s(i) = s(j) = \varnothing, \\ 0, & \text{otherwise}, \end{cases}$$

where p_i is the patch's backprojected 3D point and σ is set per dataset (0.06–0.125 m).
A query assigned to a region attends only within that region. An unassigned query attends to summaries of the regions and to nearby unassigned cells, weighted by the 3D locality bias.



Regular vs. restricted attention for a surface (table) query. Notice how attention moves off the vase and flower and completely to the table.

Step 3: choose where the final block runs. Per frame: each frame's final block runs with the restriction, and the resulting features are fused into voxels. In 3D: we fuse pre-final block input features into voxels, associate regions across frames into 3D objects, and apply the same restriction at query time. The scope therefore spans all views of an object, and the last block of attention runs once on the map at query-time. Scopes larger than 1024 voxels are randomly subsampled at query time, and unassigned voxels attend within their 0.25 m neighborhood.

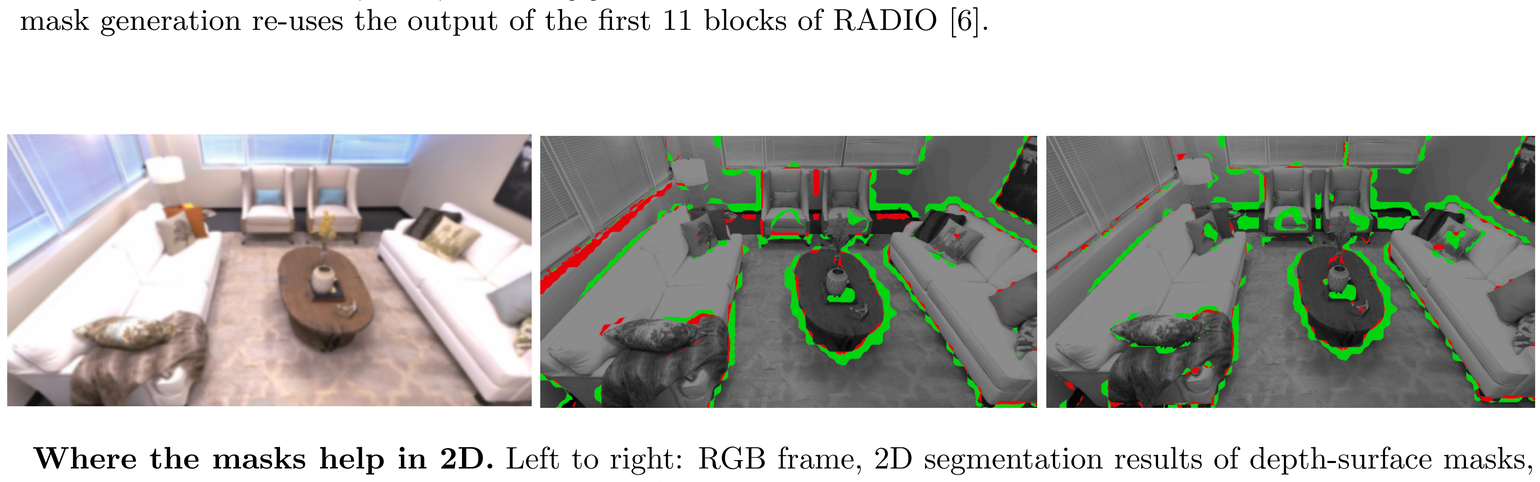


On the left, we can see how the query patch can only attend in 2D to a sliver of the stool. On the right, we can see how 3D attention affects the voxel associated with that patch - it can attend to the whole stool.

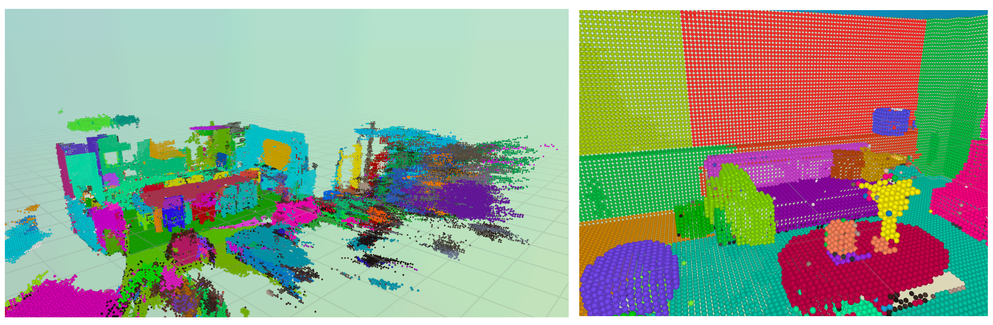
Results

Method	Regions	Attn	Replica mIoU	Replica f-mIoU	Replica FPS	ScanNet mIoU	ScanNet f-mIoU	ScanNet FPS
Baseline (RADSeg [2])	–	2D	31.5	58.4	**62.1**	38.4	47.2	**60.6**
SAM regions	SAM	2D	**34.1**	<u>61.2</u>	18.5	<u>41.0</u>	<u>51.1</u>	<u>18.4</u>
Depth surfaces	planes	2D	32.4	61.0	<u>26.0</u>	39.4	48.4	13.0
SAM objects	SAM	3D	<u>34.0</u>	**63.6**	16.0	**41.0**	**52.1**	17.7

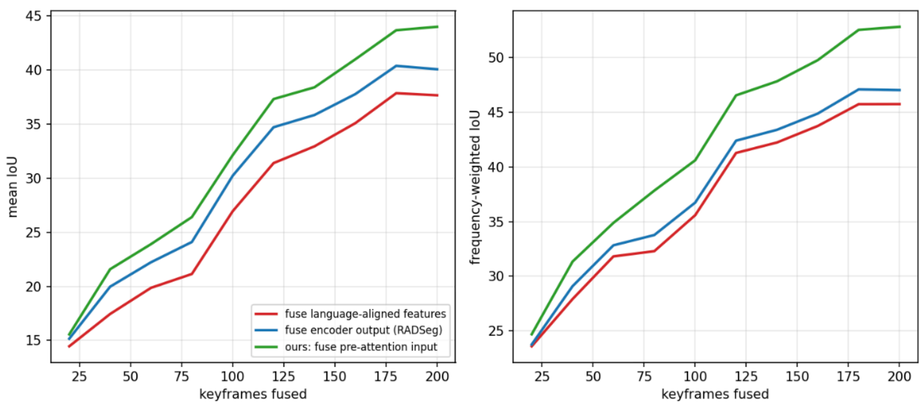
Replica (8 scenes) and ScanNet (12 scenes) at 480 × 640 on an RTX 5090. ScanNet scores in 37 classes, excluding the "other" categories, and Replica with background. Every method applies the bias at attention and SCRA. FPS covers frame time encoding, mask decoding (18 ms, SAM rows) and for ScanNet depth surfaces, plane fitting and PDA depth completion (43 ms) – PDA [4] runs an additional 12-block transformer, hence the slowdown. SAM mask generation re-uses the output of the first 11 blocks of RADIO [6].



Where the masks help in 2D. Left to right: RGB frame, 2D segmentation results of depth-surface masks, and of SAM masks. **Green** marks pixels we fix, **red** those we break when compared to encoding without the masks. Green dominates the object boundaries, where restricted attention stops features from bleeding across surfaces. Note the window sill and pillow: SAM correctly segments this and thus improves them, while purely depth-drive surface segmentation does not.



Every voxel is the color of the 3D object used at attention time. On **ScanNet**, sensor depth is noisy, so masks from different views land on misaligned voxels and association both fragments and over-merges — only 86% of the map is assigned, with one object absorbing 13,946 wall voxels together with 914 refrigerator voxels. On **Replica**, association between frames converges to clean objects covering 98% of the map.



Below: the gain from deferring grows with views. Evaluated every 20 keyframes over 5 ScanNet scenes. Fusing language-aligned features (red) trails fusing encoder features (blue) in mean IoU. Deferring the final attention block into the map (green) widens its lead as views accumulate.

Next step: try to improve association to be more noise resilient.

[1] D. Zhang, F. Liu, Q. Tang. CorrCLIP: Reconstructing Patch Correlations in CLIP for Open-Vocabulary Semantic Segmentation. *ICCV*, 2025. [2] O. Alama, D. Jariwala, A. Bhattacharya, S. Kim, W. Wang, S. Scherer. RADSeg: Unleashing Parameter and Compute Efficient Zero-Shot Open-Vocabulary Segmentation Using Agglomerative Models. *CVPR Findings*, 2026. [3] A. Kirillov et al. Segment Anything. *ICCV*, 2023. [4] Z. Wang et al. Depth Anything with Any Prior. *arXiv:2505.10565*, 2025. [5] P. F. Proença, Y. Gao. Fast Cylinder and Plane Extraction from Depth Cameras for Visual Odometry. *IROS*, 2018. [6] G. Heinrich, M. Ranzinger, H. Yin, Y. Lu, J. Kautz, A. Tao, B. Catanzaro, P. Molchanov. RADIOv2.5: Improved Baselines for Agglomerative Vision Foundation Models. *CVPR*, 2025.